IV Bolus Practice 2

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

a 250 mg of an antibiotic was administered as an IV bolus injection to a 70-kg patient. The concentration at 4 and 8 hours after administration of the dose was 6.6 mcg/mL and 2.18 mcg/mL, respectively. What is the elimination rate constant?

b An antibiotic given as a 250 mg IV bolus injection has an elimination rate constant of 0.2769 hr^{-1}. What is its half-life of elimination?

c An antibiotic given as a 250 mg IV bolus injection has an elimination rate constant of 0.2769 hr^{-1} and a plasma concentration of 2.18 mcg/mL at 8 hours. What is the initial plasma concentration?

d An antibiotic given as a 250 mg IV bolus injection has an initial plasma concentration of 20 mg/L and an elimination rate constant of 0.2769 hr^{-1}. What is the concentration of drug in the plasma 15 minutes after the dose was given?

e A 250 mg intravenous bolus dose of an antibiotic produces an initial plasma concentration of 20 mg/L in a 70-kg patient. What is the apparent volume of distribution?

f An antibiotic has an apparent volume of distribution of 12.5 L and an elimination rate constant of 0.2769 hr^{-1} in a patient. What is the total body clearance of this drug in this patient?

g An antibiotic was given as a 250 mg IV bolus injection and has an elimination rate constant of 0.2769 hr^{-1}. What is the amount of drug in the body 3 hours after the drug was administered?

h An antibiotic given by intravenous bolus injection has an elimination half-life of 2.5 hours. How much time is required for 99.9% of the drug to be eliminated from the body?